

Supplementary information for the paper: “Behind the re-rank: Explaining Fairness in Top-N Recommendations”, submitted to UMAP 2026

Eight scenarios used in the user study.

F1Q1



You have provided the following ratings about your previous visited restaurants:

Restaurant name	Restaurant A	Restaurant B	Restaurant C	Restaurant D	Restaurant E
Restaurant attributes	(Indian, fusion)	(Mexican, fast casual)	(Local cuisine, traditional)	(Japanese, sushi)	(Local cuisine, fast casual)
Rating	5	4	4	5	4

The recommender systems have suggested you the following restaurants, sorted by tentative preferences:

Position	Restaurant name	Restaurant attributes
1	Restaurant T	(Indian)
2	Restaurant U	(Local cuisine)
3	Restaurant V	(Mediterranean, fast casual)
4	Restaurant W	(Local cuisine, fast casual)
5	Restaurant X	(Japanese, sushi)
6	Restaurant Y	(Burger, Fast casual)
7	Restaurant Z	(Mexican)

The system provides the following explanation on *how* and *why* it boosted Local Cuisine in this specific list:

"To promote Local Cuisine, this recommendation list was generated by acting as if you hadn't rated **Restaurant D**"

Please, evaluate the following claims using a 5-point Likert scale ranging from 1 (completely disagree) to 5 (completely agree).

	(1) Completely disagree	(2) Somewhat disagree	(3) Neither agree nor disagree	(4) Somewhat agree	(5) Completely agree
This explanation makes me confident that I will like these restaurants.	☐	☐	☐	☐	☐
This explanation makes the recommendation process clear to me.	☐	☐	☐	☐	☐
I would enjoy using a recommendation system if it presented recommendations in this way.	☐	☐	☐	☐	☐
This explanation for the recommendation is convincing.	☐	☐	☐	☐	☐

F2Q2



You have provided the following ratings about your previous visited restaurants:

Restaurant name	Restaurant A	Restaurant B	Restaurant C	Restaurant D	Restaurant E
Restaurant attributes	(Indian, fusion)	(Mexican, fast casual)	(Local cuisine, traditional)	(Japanese, sushi)	(Local cuisine, fast casual)
Rating	5	4	4	5	4

The recommender systems have suggested you the following restaurants, sorted by tentative preferences:

Position	Restaurant name	Restaurant attributes
1	Restaurant T	(Indian)
2	Restaurant U	(Local cuisine)
3	Restaurant V	(Mediterranean, fast casual)
4	Restaurant W	(Local cuisine, fast casual)
5	Restaurant X	(Japanese, sushi)
6	Restaurant Y	(Burger, Fast casual)
7	Restaurant Z	(Mexican)

The system provides the following explanation on *how* and *why* it boosted Local Cuisine in this specific list:

"To promote Local Cuisine, this recommendation list was generated by acting as if you hadn't rated **Restaurant B**"

Please, evaluate the following claims using a 5-point Likert scale ranging from 1 (completely disagree) to 5 (completely agree).

	(1) Completely disagree	(2) Somewhat disagree	(3) Neither agree nor disagree	(4) Somewhat agree	(5) Completely agree
This explanation makes me confident that I will like these restaurants.	☐	☐	☐	☐	☐
This explanation makes the recommendation process clear to me.	☐	☐	☐	☐	☐
I would enjoy using a recommendation system if it presented recommendations in this way.	☐	☐	☐	☐	☐
This explanation for the recommendation is convincing.	☐	☐	☐	☐	☐

You have provided the following ratings about your previous visited restaurants:

Restaurant name	Restaurant A	Restaurant B	Restaurant C	Restaurant D	Restaurant E
Restaurant attributes	(Indian, fusion)	(Mexican, fast casual)	(Local cuisine, traditional)	(Japanese, sushi)	(Local cuisine, fast casual)
Rating	5	4	4	5	4

The recommender systems have suggested you the following restaurants, sorted by tentative preferences:

Position	Restaurant name	Restaurant attributes
1	Restaurant T	(Indian)
2	Restaurant U	(Local cuisine)
3	Restaurant V	(Mediterranean, fast casual)
4	Restaurant W	(Local cuisine, fast casual)
5	Restaurant X	(Japanese, sushi)
6	Restaurant Y	(Burger, Fast casual)
7	Restaurant Z	(Mexican)

The system provides the following explanation on *how* and *why* it boosted Local Cuisine in this specific list:

"Globally, your strong preference for **Mexican food** limited how much we could boost Local Cuisine in this recommendation list."

Please, evaluate the following claims using a 5-point Likert scale ranging from 1 (completely disagree) to 5 (completely agree).

	(1) Completely disagree	(2) Somewhat disagree	(3) Neither agree nor disagree	(4) Somewhat agree	(5) Completely agree
This explanation makes me confident that I will like these restaurants.	☐	☐	☐	☐	☐
This explanation makes the recommendation process clear to me.	☐	☐	☐	☐	☐
I would enjoy using a recommendation system if it presented recommendations in this way.	☐	☐	☐	☐	☐
This explanation for the recommendation is convincing.	☐	☐	☐	☐	☐

You have provided the following ratings about your previous visited restaurants:

Restaurant name	Restaurant A	Restaurant B	Restaurant C	Restaurant D	Restaurant E
Restaurant attributes	(Indian, fusion)	(Mexican, fast casual)	(Local cuisine, traditional)	(Japanese, sushi)	(Local cuisine, fast casual)
Rating	5	4	4	5	4

The recommender systems have suggested you the following restaurants, sorted by tentative preferences:

Position	Restaurant name	Restaurant attributes
1	Restaurant T	(Indian)
2	Restaurant U	(Local cuisine)
3	Restaurant V	(Mediterranean, fast casual)
4	Restaurant W	(Local cuisine, fast casual)
5	Restaurant X	(Japanese, sushi)
6	Restaurant Y	(Burger, Fast casual)
7	Restaurant Z	(Mexican)

The system provides the following explanation on *how* and *why* it boosted Local Cuisine in this specific list:

"Globally, your strong preference for **Japanese food** limited how much we could boost Local Cuisine in this recommendation list."

Please, evaluate the following claims using a 5-point Likert scale ranging from 1 (completely disagree) to 5 (completely agree).

	(1) Completely disagree	(2) Somewhat disagree	(3) Neither agree nor disagree	(4) Somewhat agree	(5) Completely agree
This explanation makes me confident that I will like these restaurants.	☐	☐	☐	☐	☐
This explanation makes the recommendation process clear to me.	☐	☐	☐	☐	☐
I would enjoy using a recommendation system if it presented recommendations in this way.	☐	☐	☐	☐	☐
This explanation for the recommendation is convincing.	☐	☐	☐	☐	☐

F5Q5



You have provided the following ratings about your previous visited restaurants:

Restaurant name	Restaurant A	Restaurant B	Restaurant C	Restaurant D	Restaurant E
Restaurant attributes	(Indian, fusion)	(Mexican, fast casual)	(Local cuisine, traditional)	(Japanese, sushi)	(Local cuisine, fast casual)
Rating	5	4	4	5	4

The recommender systems have suggested you the following restaurants, sorted by tentative preferences:

Position	Restaurant name	Restaurant attributes
1	Restaurant T	(Indian)
2	Restaurant U	(Local cuisine)
3	Restaurant V	(Mediterranean, fast casual)
4	Restaurant W	(Local cuisine, fast casual)
5	Restaurant X	(Japanese, sushi)
6	Restaurant Y	(Burger, Fast casual)
7	Restaurant Z	(Mexican)

The system provides the following explanation on *how* and *why* it boosted Local Cuisine in this specific list:

"To boost Local Cuisine, we specifically lowered the rank of **Mexican 'Fast Casual'** restaurants from your recommendations.."

Please, evaluate the following claims using a 5-point Likert scale ranging from 1 (completely disagree) to 5 (completely agree).

	(1) Completely disagree	(2) Somewhat disagree	(3) Neither agree nor disagree	(4) Somewhat agree	(5) Completely agree
This explanation makes me confident that I will like these restaurants.	☐	☐	☐	☐	☐
This explanation makes the recommendation process clear to me.	☐	☐	☐	☐	☐
I would enjoy using a recommendation system if it presented recommendations in this way.	☐	☐	☐	☐	☐
This explanation for the recommendation is convincing.	☐	☐	☐	☐	☐

F6Q6



You have provided the following ratings about your previous visited restaurants:

Restaurant name	Restaurant A	Restaurant B	Restaurant C	Restaurant D	Restaurant E
Restaurant attributes	(Indian, fusion)	(Mexican, fast casual)	(Local cuisine, traditional)	(Japanese, sushi)	(Local cuisine, fast casual)
Rating	5	4	4	5	4

The recommender systems have suggested you the following restaurants, sorted by tentative preferences:

Position	Restaurant name	Restaurant attributes
1	Restaurant T	(Indian)
2	Restaurant U	(Local cuisine)
3	Restaurant V	(Mediterranean, fast casual)
4	Restaurant W	(Local cuisine, fast casual)
5	Restaurant X	(Japanese, sushi)
6	Restaurant Y	(Burger, Fast casual)
7	Restaurant Z	(Mexican)

The system provides the following explanation on *how* and *why* it boosted Local Cuisine in this specific list:

"To boost Local Cuisine, we specifically lowered the rank of **Japanese 'Sushi'** restaurants from your recommendations.."

Please, evaluate the following claims using a 5-point Likert scale ranging from 1 (completely disagree) to 5 (completely agree).

	(1) Completely disagree	(2) Somewhat disagree	(3) Neither agree nor disagree	(4) Somewhat agree	(5) Completely agree
This explanation makes me confident that I will like these restaurants.	☐	☐	☐	☐	☐
This explanation makes the recommendation process clear to me.	☐	☐	☐	☐	☐
I would enjoy using a recommendation system if it presented recommendations in this way.	☐	☐	☐	☐	☐
This explanation for the recommendation is convincing.	☐	☐	☐	☐	☐

F7Q7



You have provided the following ratings about your previous visited restaurants:

Restaurant name	Restaurant A	Restaurant B	Restaurant C	Restaurant D	Restaurant E
Restaurant attributes	(Indian, fusion)	(Mexican, fast casual)	(Local cuisine, traditional)	(Japanese, sushi)	(Local cuisine, fast casual)
Rating	5	4	4	5	4

The recommender systems have suggested you the following restaurants, sorted by tentative preferences:

Position	Restaurant name	Restaurant attributes
1	Restaurant T	(Indian)
2	Restaurant U	(Local cuisine)
3	Restaurant V	(Mediterranean, fast casual)
4	Restaurant W	(Local cuisine, fast casual)
5	Restaurant X	(Japanese, sushi)
6	Restaurant Y	(Burger, Fast casual)
7	Restaurant Z	(Mexican)

The system provides the following explanation on *how* and *why* it boosted Local Cuisine in this specific list:

"Focusing on Local Cuisine had the side effect of promoting **Burger restaurants** as a similar alternative to your usual spots."

Please, evaluate the following claims using a 5-point Likert scale ranging from 1 (completely disagree) to 5 (completely agree).

	(1) Completely disagree	(2) Somewhat disagree	(3) Neither agree nor disagree	(4) Somewhat agree	(5) Completely agree
This explanation makes me confident that I will like these restaurants.	☐	☐	☐	☐	☐
This explanation makes the recommendation process clear to me.	☐	☐	☐	☐	☐
I would enjoy using a recommendation system if it presented recommendations in this way.	☐	☐	☐	☐	☐
This explanation for the recommendation is convincing.	☐	☐	☐	☐	☐

F8Q8



You have provided the following ratings about your previous visited restaurants:

Restaurant name	Restaurant A	Restaurant B	Restaurant C	Restaurant D	Restaurant E
Restaurant attributes	(Indian, fusion)	(Mexican, fast casual)	(Local cuisine, traditional)	(Japanese, sushi)	(Local cuisine, fast casual)
Rating	5	4	4	5	4

The recommender systems have suggested you the following restaurants, sorted by tentative preferences:

Position	Restaurant name	Restaurant attributes
1	Restaurant T	(Indian)
2	Restaurant U	(Local cuisine)
3	Restaurant V	(Mediterranean, fast casual)
4	Restaurant W	(Local cuisine, fast casual)
5	Restaurant X	(Japanese, sushi)
6	Restaurant Y	(Burger, Fast casual)
7	Restaurant Z	(Mexican)

The system provides the following explanation on *how* and *why* it boosted Local Cuisine in this specific list:

"Focusing on Local Cuisine had the side effect of promoting **Mediterranean restaurants** as a similar alternative to your usual spots."

Please, evaluate the following claims using a 5-point Likert scale ranging from 1 (completely disagree) to 5 (completely agree).

	(1) Completely disagree	(2) Somewhat disagree	(3) Neither agree nor disagree	(4) Somewhat agree	(5) Completely agree
This explanation makes me confident that I will like these restaurants.	☐	☐	☐	☐	☐
This explanation makes the recommendation process clear to me.	☐	☐	☐	☐	☐
I would enjoy using a recommendation system if it presented recommendations in this way.	☐	☐	☐	☐	☐
This explanation for the recommendation is convincing.	☐	☐	☐	☐	☐